Robust County-Level Corn Yield Estimation Using Ensemble Machine Learning and Multisource Remote Sensing

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***Abstract*—Advancement of sustainable agriculture techniques and ensuring food security depend on accurate and reliable crop yield forecasting. Accurate and reliable crop yield forecasting is essential for sustainable agricultural planning and global food security. However, data quality issues—such as missing values and temporal misalignments in remote sensing datasets—often challenge the robustness of machine learning models. This study proposes a robust yield prediction framework that integrates multisource data, including MODIS-based gross primary production (GPP), vegetation indices (NDVI, EVI), climate variables, and soil properties, to estimate maize yield at the county level across the U.S. Corn Belt. Two ensemble models, random forest (RF) and extreme gradient boosting (XGBoost), are trained and evaluated under both clean and simulated degraded data conditions. XGBoost achieved the highest accuracy (RMSE = 14.58, $R^2 = 0.84$), while RF demonstrated strong robustness (RMSE = 15.10, $R^2 = 0.82$), even when early-season NDVI was missing or GPP time series were temporally shifted. Feature importance analysis identified late-season GPP and soil organic matter as the most influential predictors. These findings demonstrate the potential of ensemble learning models to deliver reliable and interpretable yield forecasts, even under imperfect data conditions, making them practical tools for real-world precision agriculture applications. This work provides a practical framework for implementing artificial intelligence-driven solutions in large-scale, real-world remote sensing-based agricultural monitoring, and yield forecasting systems.**

***Index Terms*—AI robustness, crop yield prediction, data quality, extreme gradient boosting (XGBoost), multisource data fusion, precision agriculture, random forest, remote sensing.**

I. Introduction

THE integration of artificial intelligence (AI) and remote sensing has significantly advanced land management, precision agriculture, and environmental monitoring by enabling data-driven insights and scalable decision-making [1]. Among these uses, crop yield prediction is clearly a high-impact one that supports choices in food security, supply chain management, and climate resilience [2]. Thanks to growing access to multispectral and multisource remote sensing data—from satellites, drones, and ground-based sensors—researchers may now simulate biophysical processes at unprecedented spatial and temporal resolutions [3], [4]. But the quality and consistency of the input data determine the dependability of AI models created on these datasets [5], [6].

Precision agriculture plays a critical role in ensuring food security, enhancing crop productivity, and promoting sustainable land management in the face of global climate challenges. To achieve these goals, the agricultural sector increasingly relies on data-driven technologies to monitor crop conditions, assess environmental variables, and optimize decision-making. Among these technologies, the integration of AI and remote sensing has emerged as a transformative approach. By leveraging satellite imagery, vegetation indices, and machine learning models, researchers can now generate accurate and timely insights at unprecedented spatial and temporal resolutions—enabling predictive tools for yield forecasting and resource management.

Remote sensing data by nature is prone to several quality problems [7]. These encompass ambient noise, cloud cover, sensor calibration mistakes, temporal misalignment across data sources, and missing values brought on by platform-specific restrictions or acquisition failures [8]. Such flaws can greatly reduce the generalizability of predictions, cause biases, and seriously compromise model performance [9]. Though a lot of research on machine learning applications in remote sensing is under progress, very few studies have methodically assessed how often data quality problems compromise the interpretability and resilience of AI models in useful environments [4], [10].

This work fills up this gap by suggesting a strong machine learning framework for agricultural production prediction under simulated data degradation conditions. We integrate time-series gross primary production (GPP) from MODIS satellite products [11], vegetation indices including the Normalized Difference Vegetation Index (NDVI), climate variables (e.g., temperature, precipitation, vapor pressure deficit) [12], [13], and soil properties (e.g., cation exchange capacity, organic matter content) [14], [15] using a rich dataset of county-level maize yield in the United States. Two ensemble-based regression models—random forest (RF) [16], [17] and Extreme Gradient Boosting (XGBoost) [18], [19]—which are assessed in both standard and degraded data environments—are trained using these characteristics [2], [20], [21].

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Two typical quality problems are introduced to replicate real-world challenges in remote sensing: 1) randomly simulated missing data in NDVI values to mimic data loss from cloud contamination or sensor dropout, and 2) temporal misalignment in GPP time series by artificially shifting the features by one timestep, so simulating sensor synchronization errors. With a root-mean-squared error (RMSE) of 15.10 and a R^2 score of 0.82 under normal conditions, the models—especially RF—show strong prediction accuracy even under these perturbations, hence minimizing performance decline under simulated noise and loss.

Data degradation in remote sensing is a persistent challenge that can significantly affect downstream analysis and model performance. Common issues include cloud contamination, atmospheric noise, inconsistent sensor calibration, temporal gaps, and missing observations, all of which can introduce uncertainty and bias into predictive models. Several recent studies have specifically addressed the implications of such degradation on the quality and reliability of remote sensing products [22], [23]. For example, cloud and shadow masking remains a major obstacle for optical satellite imagery [22], while temporal misalignment between different sensor acquisitions can lead to mismatches in multisource data fusion [24]. Moreover, approaches for handling missing data—such as gap-filling algorithms and imputation techniques—have been reviewed in the context of remote sensing time series [23], [25]. Recognizing and mitigating these data quality issues is essential for the development of robust machine learning models in operational remote sensing applications.

Moreover, this work offers a thorough investigation of feature significance, residual distribution, and model interpretability in order to grasp, which factors have the most impact and how the prediction uncertainty of the model is dispersed. The results emphasize the main influence of late-season GPP (more especially GPP_8) and soil organic matter in influencing crop output, which fits agronomic knowledge on maize phenology and soil fertility dynamics.

This study stands out for not only creating a high-performance AI-based yield estimate framework but also for extensively assessing the sensitivity of the model to reasonable data flaws. We systematically assess the prediction model stability and dependability by means of controlled simulations with missing data and sensor misalignment. This double emphasis on resilience and accuracy provides a whole methodology for remote sensing-based agricultural analytics that mirrors actual operational difficulties. In precision agriculture, where consistent data quality cannot always be assured, the findings shown in this work represent a significant step forward in bridging the gap between theoretical model development and actual application. In the end, our work prepares the way for creating AI systems that are not only accurate but also strong, interpretable, and deployable in many and flawed sensing settings.

The rest of this article is organized as follows. Section II presents a review of related work in crop yield prediction using remote sensing and machine learning. Section III describes the materials and methods, including the study area, datasets, and model development. Section IV provides an overview of the evaluation metrics used to assess model performance. Also, this section discusses the experimental results, feature importance, and robustness analysis. Finally, Section V concludes this article.

II. Related Works-State of Art

Especially in the field of crop yield estimate, the junction of AI with remote sensing has resulted in major developments in agricultural monitoring and prediction [26]. Researchers have investigated a broad spectrum of machine learning methods to extract significant insights from satellite-derived indicators as data access from Earth observation platforms has grown [3], [27]. Though these research show the predictive capabilities of AI-based models, very few have methodically investigated how model resilience and operational dependability vary with the quality of input data [28], [29].

In recent years, the application of machine learning in remote sensing for agricultural productivity modeling has gained significant momentum. Crop yield estimation using satellite-derived indicators such as the NDVI and GPP has been widely explored due to their proven correlation with photosynthetic activity and biomass accumulation. Johnson [27] demonstrated that multitemporal NDVI from MODIS could provide accurate yield estimates across diverse agroecological zones. Similarly, Lyu et al. [30] integrated MODIS GPP with historical yield records and achieved strong predictive performance using RF regression models.

Advanced machine learning algorithms such as RFs and XGBoost have become popular due to their robustness to overfitting and capacity to handle high-dimensional, nonlinear relationships. Studies by You et al. [31] and Khaki and Wang [32] successfully applied these ensemble learning models to estimate yields using time-series vegetation indices and climate variables, showing substantial improvements over traditional statistical methods.

However, fewer studies have explicitly addressed the impact of data quality degradation—such as missing values or misaligned time series—on model reliability. Wang et al. [33] explored the effects of cloud-induced data gaps on crop classification performance but focused primarily on remote sensing preprocessing rather than downstream AI robustness. More recently, Ferchichi et al. [34] highlighted the vulnerability of deep learning models to noise in satellite imagery, calling for new methods to quantify and mitigate uncertainty in AI-driven remote sensing applications.

In the context of precision agriculture, the integration of soil properties with remotely sensed features has shown to significantly improve yield estimation. Lobell et al. [29] combined satellite data with *in situ* soil measurements and demonstrated that models incorporating both sources are more generalizable across regions. Nonetheless, even these integrated approaches rarely simulate real-world disturbances in sensor data, which are common in operational systems.

Lyu et al. [30] employed time-series MODIS data with RF models and interpretability tools for maize yield prediction, demonstrating improved performance over static models. Li et al. [18] proposed a county-level soybean yield prediction framework using XGBoost and multidimensional feature engineering, showcasing high scalability. Additionally, Ferchichi et al. [34] provided a comprehensive review on uncertainty quantification in deep learning models for remote sensing, underlining the sensitivity of predictions to noise. More recently, Zhou et al. [26]

assessed whether improving data quality or increasing model complexity contributes more significantly to yield forecasting accuracy, offering valuable insights into practical model design. These studies reinforce the relevance of ensemble and interpretable models—such as those used in this work—for robust, real-world agricultural applications.

This study builds upon and extends these prior works by combining high-resolution, multisource datasets and rigorously evaluating model performance under controlled data degradation scenarios. By simulating missing vegetation index data and GPP misalignment, we aim to bridge the gap between idealized AI model development and practical remote sensing deployment, contributing novel insights into the resilience and interpretability of ensemble models under noisy conditions.

In view of these limitations, the present work provides a unique contribution by directly replicating two typical data quality challenges: temporal misalignment of GPP signals and missing vegetation index values. We give a fresh evaluation of model resilience in precision agriculture by assessing RF and XGBoost models under these degraded settings. This method provides understanding of how data defects affect both predicted accuracy and feature interpretation, hence bridging the gap between controlled model building and the erratic character of operational remote sensing.

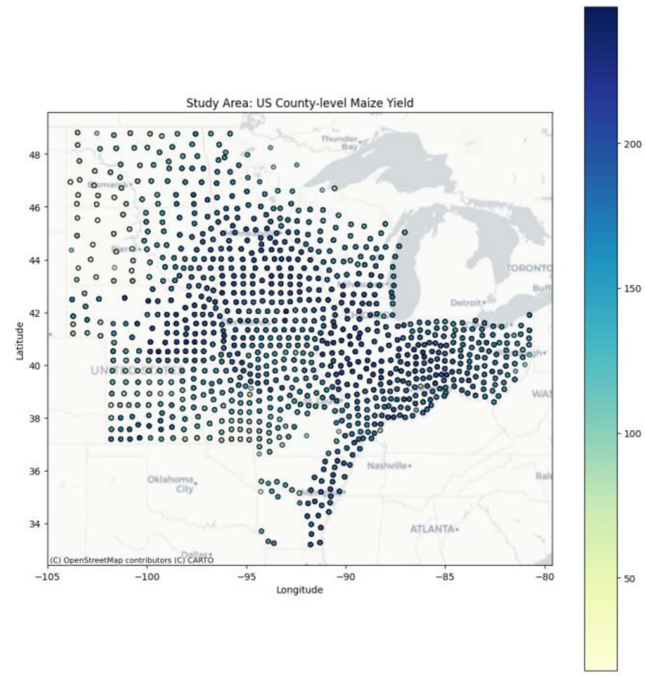

Fig. 1. Spatial distribution of the study area showing county-level maize yield across the U.S. Midwest and surrounding regions. Each point represents the centroid of a county, with color intensity indicating maize yield (bushels per acre).

III. Material and Methods

This work used a complete approach combining multisource remote sensing and environmental datasets with machine learning models to evaluate how data quality problems affect AI-based agricultural yield prediction. Entire feature set containing satellite-derived GPP, vegetation indices (NDVI and EVI), climatic variables, and soil parameters trained ensemble regression algorithms. The approach has been developed to measure resilience in the presence of reasonable constraints including missing data and temporal misalignment in addition to baseline model performance under perfect data conditions. Strong generalization capacity and resistance to overfitting in high-dimensional feature spaces drove two well-known ensemble learning techniques, RF and XGBoost, to be chosen. These models' performance has assessed using conventional evaluation criteria and subsequently examined by means of feature significance and residual distribution analysis. The dataset, preparation procedures, experimental design, and robustness simulations in particular are covered in the following sections.

A. Study Area and Dataset

Using a high-dimensional dataset including satellite-derived biophysical indicators, climatic factors, and soil properties [29], [35], this work focuses on maize yield prediction across United States counties, as shown in Fig. 1. The study's spatial scope spans many U.S. states and counties having accessible annual agricultural production records, therefore offering a geographically varied testbed for assessing model generality. With each record reflecting a county-level observation for a given year, the temporal sweep encompasses many expanding seasons.

The study area encompasses the U.S. Corn Belt, a key agricultural region that includes major maize-producing states such as Iowa, Illinois, Indiana, Nebraska, and Minnesota. This region is characterized by fertile soils, moderate to high precipitation levels during the growing season, and well-established infrastructure for row-crop agriculture. The dataset includes county-level maize yield records spanning multiple years, offering spatial and temporal heterogeneity necessary for robust model evaluation. Fig. 1 presents the spatial distribution of the study area, showing the geographic extent, county centroids, and yield intensity in bushels per acre.

Measured in bushels per acre, maize yield—derived from USDA statistics—is the main aim variable. For predictor variables, the dataset combines a spectrum of properties taken from environmental sources and Earth observation. Derived from the MODIS GPP product, time-series GPP statistics are presented at 11 intervals annually, therefore offering biweekly or monthly observations across the growing season. To get real GPP figures, these numbers are scaled by a factor of 1 000 000. Using 12 time-series values each of the NDVI and enhanced vegetation index (EVI), corresponding to the crop growth phases between April and November, vegetation dynamics are recorded [20], [36], [37].

The dataset comprises variables including precipitation (PPT), mean, minimum, and maximum temperatures (TMEAN, TMIN, TMAX), vapor pressure deficit (VPDMIN, VPDMAX), and mean dew point temperature (TDMEAN), aggregated at monthly intervals to help to explain climatic impacts on yield. To reflect the edaphic elements affecting crop development, also included are thorough soil properties including organic matter content, cation exchange capacity (CEC), bulk density, accessible water content (AWC), and percentages of sand and clay.

Using a unique five-digit GEOID, each record is georeferenced combining state and county codes with geographical coordinates (X, Y) matching the county centroid. The dataset has been preprocessed to eliminate header duplicates, translate all features to numerical form, and scale the GPP variables before model training. This rich mix of temporal, spectral, and environmental data lets one evaluate machine learning models under both ideal and deteriorated settings holistically.

B. Data Preprocessing

Remote sensing-based machine learning applications depend on effective preprocessing, especially in managing heterogeneous, multisource environmental information. The preparation approach for this work handled formatting errors, scaled temporal variables, maintained variability for robustness testing, and guaranteed spatial coherence over all records. The final dataset has put up to facilitate not just baseline model training but also thorough assessment under simulated data deterioration scenarios. All data preprocessing steps—including cleaning, formatting, scaling, and geospatial alignment—were implemented using Python (version 3.9) with libraries such as pandas, NumPy, and GeoPandas.

The aerial imagery used for this study was obtained from the U.S. National Agriculture Imagery Program, which is publicly available under an open-use license. Manual digitization of cadastral boundaries was performed by trained GIS analysts using high-resolution orthoimagery in QGIS 3.28. Polygons were drawn following visible field boundaries, checked for topological consistency, and saved in GeoPackage format. Metadata about the digitization workflow and QC steps are included in the GitHub repository linked above.

The original dataset included heterogeneous data types, inconsistent column names, and duplicate header rows. The column headings are standardized and all feature values have been translated to suitable numerical forms to help to fix these problems. Entries lacking fundamental values—such as geographic coordinates or yield data—are eliminated; partial missingness in predictor variables has been kept for use in robustness simulations. This method maintained the integrity of the dataset and permitted a reasonable evaluation of model performance under limited data conditions.

Key feature categories comprising the dataset are agricultural yield, vegetation indices, productivity measures, climatic variables, soil characteristics, and geographic identifiers. Table I summarizes these several groups, their historical features, and data sources together. Target variable is maize yield, expressed in bushels per acre at the county level using USDA statistics.

The NDVI and EVI, both derived from satellites and shown as 12 separate variables (NDVI_1 to NDVI_12) monthly from April to November. Likewise, generated from the MODIS GPP product, GPP data included 11 periods (GPP_1 to GPP_11) over the expanding season. Originally in digital form, these GPP data have been scaled—i.e., adjusted to ensure values fall within a comparable numerical range—and then divided by 10 000 to convert them into physiologically interpretable units (e.g., gC/m^2/day).

TABLE I
SUMMARY OF DATASET VARIABLES AND SOURCES

Feature Group	Variable Examples	Temporal Resolution	Source	Description
Crop Yield	yield	Annual	USDA Statistics	Target variable: maize yield (bushels/acre)
Vegetation Index	NDVI_1 to NDVI_12	Monthly	MODIS	Normalized Difference Vegetation Index
Enhanced Index	EVI_1 to EVI_12	Monthly	MODIS	Enhanced Vegetation Index
Productivity	GPP_1 to GPP_11	Bi-weekly/Monthly	MODIS GPP	Gross Primary Production (scaled)
Climate	PPT_1, TMEAN_1, etc.	Monthly	NOAA/ Daymet	Precipitation, temperature, vapor pressure, etc.
Soil	organic_matter, CEC	Static	SSURGO Soil Database	Soil fertility and water capacity attributes
Spatial Metadata	GEOID, X, Y	Static	County-level shapefiles	County identifier and centroid coordinates

Reflecting important environmental influences of crop growth, climate characteristics included monthly summaries of precipitation, temperature (mean, minimum, and maximum), and vapor pressure deficit. To retain phenological tendencies, each has set up as a time series—i.e., PPT_1 through PPT_11, TMEAN_1 through TMEAN_11. Static soil variables derived from the SSURGO soil database Known to have a major impact on crop output are these: organic matter, clay percentage, AWC, and CEC. In this study, "NDVI_1" denotes the Normalized Difference Vegetation Index value acquired in April, corresponding to the early vegetative growth stage of maize in the U.S. Corn Belt. Subsequent NDVI variables (NDVI_2 to NDVI_12) represent monthly observations from May through November, aligning with key phenological phases of the crop

Although exploratory methods (e.g., correlation matrices, feature significance plots) temporarily used z-score normalizing to permit interpretability across variables with varied units and scales, machine learning models such as RF and XGBoost do not require input standardization. Every data is geospatially linked using GEOID, a five-digit identification comprising state and county codes, as well as geographic coordinates (X, Y) indicating the centroid of every county. This spatial tagging guaranteed regional consistency in model evaluation and permitted visual mapping of yield patterns—as illustrated in Fig. 1—to be possible. We selected missing NDVI_1 and a one-timestep shift in GPP to reflect common data issues in remote sensing. Early-season NDVI is often missing due to cloud cover or acquisition gaps, while slight temporal misalignments in GPP can result from differences in satellite revisit schedules. The chosen magnitudes are representative of typical challenges encountered in operational agricultural monitoring.

To evaluate model robustness under real-world data quality issues, two types of data degradation were simulated during preprocessing. First, missing data were introduced by randomly removing values from the NDVI_1 variable (representing early-season vegetation conditions). Approximately 10% of the values in this feature were masked to emulate missing observations due to cloud contamination, sensor dropout, or data transmission errors. Second, temporal misalignment was simulated by shifting all GPP time series variables (GPP_1 to GPP_11) by one time step forward. This effectively introduced a lag, mimicking issues such as sensor desynchronization, delayed satellite acquisitions, or temporal mismatch with ground observations. These perturbations were applied only during specific experimental runs and compared with baseline model performance on the original dataset to assess the stability and resilience of the models.

C. RF Regression Modeling Framework

This study models maize yield with high-dimensional multi-source data using an ensemble learning method. Two models—RF regression and XGBoost—have been chosen depending on their shown resilience in handling structured tabular data and their broad use in remote sensing and environmental sciences: By means of built-in feature significance evaluation, both models are well-suited for capturing nonlinear interactions among features, thereby reducing overfitting in high-dimensional spaces, and so promoting interpretability. Independent implementation of the models allowed a direct comparison of resilience under simulated data quality concerns with prediction performance.

RF and XGBoost were selected for their proven ability to handle high-dimensional, heterogeneous datasets commonly found in remote sensing and environmental modeling. RF is an ensemble learning method based on bootstrap aggregation (bagging), which offers strong generalization performance, robustness to noise, and natural handling of feature importance via Gini impurity reduction. XGBoost, on the other hand, is based on gradient boosting and is designed to iteratively correct weak learners, achieving high predictive accuracy through regularization, parallelization, and tree pruning mechanisms. Both models are well-suited to the nonlinear relationships and multicollinearity present in our multisource dataset. While RF is computationally more efficient and less sensitive to overfitting, XGBoost tends to offer higher precision but is more complex to tune and slightly more vulnerable to noisy or missing input data. These complementary characteristics make them ideal for comparative evaluation in the context of yield prediction under real-world data imperfections.

Operating on the bootstrap aggregation (bagging) idea, RF is a quite popular ensemble learning approach. It averages the outcomes of many decision trees created on randomly selected subsets of the training data to provide an ultimate forecast. This method is resilient to multicollinearity and noise in the input data, helps to naturally lower variance, and avoids overfitting.

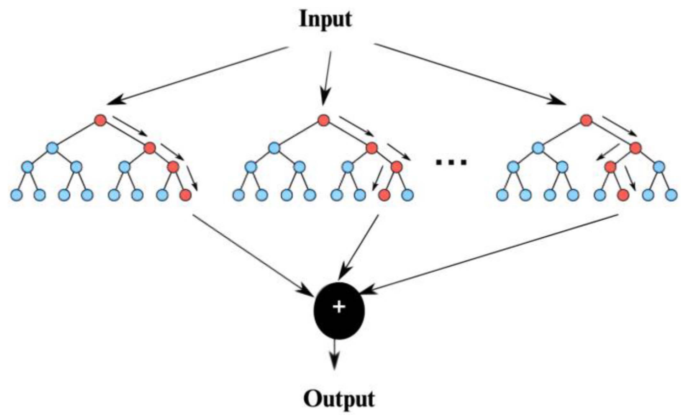

Fig. 2. Schematic representation of the RF regression architecture.

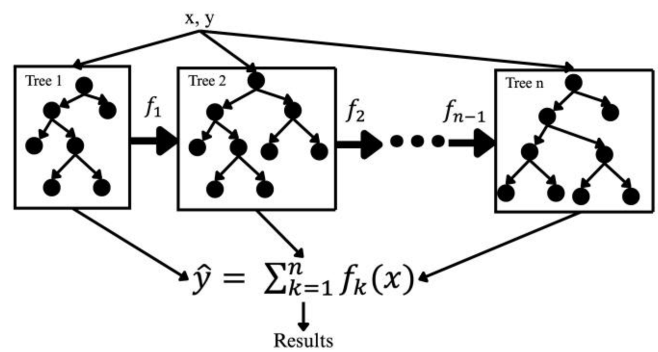

Fig. 3. Diagram of the XGBoost regression architecture.

This work implemented the RF regressor from the scikit-learn module as the RF model (see Fig. 2). To guarantee repeatability, the model has been started with one hundred decision trees (n_estimators = 100) and a set seed (random_state = 42). Although the goal variable is maize yield expressed in bushels per acre, the training dataset comprised a broad spectrum of predictors—including GPP, NDVI, EVI, climate, and soil variables. Preserving yield variability across both subsets, the dataset has randomly split into a training set (80%) and a test set (20%).

Strong baseline performance has been attained by the RF model with a RMSE of 15.10 and a coefficient of determination (R^2) of 0.82, therefore showing good prediction accuracy. Crucially, the scale-invariance of tree-based techniques allowed the model to demand no feature scaling. The Gini-based feature significance metric is then used to examine the model's output more closely since it ranks input variables according to their contribution to lower impurity across all decision nodes.

Results showed that yield forecasts are most influenced by GPP variables—particularly GPP_8, GPP_7, and GPP_6. These line up with late-season output and fall during important phenological phases like grain filling and maturity. Furthermore highly ranked are static soil parameters like organic matter and CEC, which represent the long-term impact of edaphic elements on crop output. Simulating sensor degradation scenarios—including the insertion of missing NDVI data and intentional misalignment of GPP time-series features—helps one further assess the resilience of the RF model. Under both scenarios, the model maintained steady performance, therefore verifying its robustness to modest degrees of data imperfection—a crucial factor for practical implementation when data quality cannot always be under control.

D. XGBoost Regression Architecture

Apart based on boosting, XGBoost is a potent machine learning method in which weak learners are progressively and iteratively corrected upon past mistakes. By maximizing a loss function in a gradient descent framework, boosting lowers both bias and variance unlike bagging techniques. In processing structured data, XGBoost is quite efficient and effective as it brings various improvements over conventional boosting algorithms including regularization (L1 and L2), tree pruning, and parallel computing. This effort-maintained consistency with the RF model using the XGBRegressor from the XGBoost Python library with n_estimators = 100 and random_state = 42. The two techniques may be fairly compared using the same training and testing sets as well as the same set of predictor variables applied in the model.

With very minor variances in residual distribution, the XGBoost model attained performance measures on par with RF. Although XGBoost's repeated learning method usually concentrates on the most challenging samples, occasionally it produces greater generalization for noisy or skewed data. Based on gain, cover, or weight, its feature important output verified the relevance of GPP_8, GPP_7, and organic_matter, so suggesting consistency in varied effect across both models.

Additionally assessed under the same simulated data deterioration conditions is XGBoost's performance. Although it showed great generality, its rather more complicated optimization method rendered it rather more vulnerable to NDVI data gaps than RF. Still, its performance stayed within reasonable limits, which confirmed its fit for yield prediction applications including flawed remote sensing data.

Using these two ensembles learning techniques, this work offers a strong comparative examination of prediction dependability under practical limitations. Their complementary strengths—XGBoost's bias correction and RF's variance reduction—provide insightful analysis for further implementations in remote sensing data-based agricultural monitoring systems.

E. Evaluation Metrics

Particularly in the framework of remote sensing applications where data uncertainty is widespread, a thorough assessment of machine learning models is necessary to guarantee both forecast accuracy and operational dependability. This work evaluates model performance holistically under both baseline and simulated data deterioration conditions using many assessment indicators. Average prediction error, variance explained, and model generalizability are among the several performance parameters that have been chosen to measure. A fair knowledge of the performance of the models over a wide range of environmental and spectral characteristics has obtained using both numerical and graphical methods.

The present study combined diagnostic visualization methods with traditional regression assessment measures to fully evaluate model performance under both ideal and poor data situations. These measures have been chosen depending on their applicability to remote sensing and precision agriculture, where accuracy and interpretability rule first. Resual distribution analysis, feature importance ranking, and scatter plots of actual against predicted yields complemented the two fundamental quantitative metrics—RMSE and coefficient of determination (R^2)—all of which are applied in the code script to ensure interpretability and transparency. Measuring the average size of prediction mistakes, RMSE is a commonly used statistic that gives more weight on bigger errors because of squaring. It is appropriate for evaluating the general deviation from actual values as it offers an interpretable value in the same unit as the expected variable—in this example, maize yield in bushels per acre [37]

$$\mathrm{RMSE} = \sqrt{\frac{1}{n} \sum\nolimits_{i=1}^{n} (y_i - \hat{y}_i)^2}. \tag{1}$$

In the baseline scenario, the RF model achieved an RMSE of approximately 15.10, while the XGBoost model yielded similar results, confirming the predictive strength of both ensemble methods under clean data conditions. The R^2 metric quantifies the proportion of variance in the observed yield values that is explained by the model predictions. It provides a measure of goodness-of-fit, ranging from 0 (or negative, in rare cases) to 1. Higher values indicate better explanatory power [37]

$$R^2 = 1 - \frac{\sum_{i=1}^{n} (y_i - \hat{y}_i)^2}{\sum_{i=1}^{n} (y_i - \bar{y})^2}. \tag{2}$$

Baseline R^2 values reached 0.82–0.824, indicating that the models captured the majority of yield variability across spatial and temporal domains. These plots revealed a near-normal, symmetric distribution centered around zero, which suggests the models are well-calibrated and free from strong systematic bias. Residual spread is also visually analyzed under simulated missing data (NDVI) and temporal misalignment (GPP shift) conditions, where slight broadening of the distribution has been observed, but without significant skew.

By measuring the disparity between red and near-infrared reflectance, the widely used remote sensing indicator known as the NDVI assesses vegetation health and biomass. Because of the way plant leaves are structured, healthy vegetation reflects a lot of near-infrared light and absorbs most visible light—especially red—for photosynthesis. NDVI computed as follows [38], [39]:

$$\mathrm{NDVI} = \frac{\mathrm{NIR} - \mathrm{RED}}{\mathrm{NIR} + \mathrm{RED}}. \tag{3}$$

Internal mechanisms for rating feature significance based on measures such as information gain or impurity reduction (Gini important) abound in both RF and XGBoost models. This study again identified the GPP characteristics—particularly GPP_8 and GPP_7—as the most important predictors followed by organic matter and early-season meteorological indicators. These outcomes improve the interpretability and agronomic usefulness of the models as they match known agronomic factors of yield.

This work offers a strong and multifarious performance evaluation of RF and XGBoost models by including these evaluation measures. While residual plots and feature significance interpretation provide insights on model calibration and transparency, the combined usage of RMSE, R^2, MAE, MAPE, and modified R^2 allows both absolute and relative error analysis. These instruments used together help to clarify model dependability, particularly in the context of reasonable data quality problems including missing values and temporal misalignments. This thorough assessment method guarantees that the results are practically meaningful for decision-making in precision agriculture and remote sensing-based crop monitoring in addition to statistically valid.

IV. Results and Discussion

Under both conventional circumstances and simulated data quality deterioration situations, the generated RF and XGBoost models are assessed. The experimental results are thoroughly analyzed in this part together with baseline model correctness, feature significance interpretation, residual behavior, resilience against missing data and temporal misalignments, and relative performance of the two methods. Visual diagnostics like scatter plots, residual histograms, boxplots, and feature significance charts help to complement quantitative findings thereby provide a complete knowledge of model behavior. Where relevant, the effects of data quality problems on model stability and predictive capacity are explored, therefore providing information on best practices for implementing remote sensing systems driven by AI in agricultural monitoring. This section presents and discusses the performance of the developed RF and XGBoost models under both ideal and degraded data conditions, supported by visual and statistical evaluations.

The segment anything model (SAM) used in this study was implemented using the official Meta AI release, with the ViT-Huge) checkpoint selected for its superior segmentation performance. A bounding box prompting strategy was employed, leveraging the centroids of reference cadastral polygons to initiate mask extraction. All inference and evaluation procedures were run on a system equipped with an NVIDIA RTX 3090 GPU (24 GB), 128 GB RAM, and an Intel Xeon processor, enabling fast tile-level predictions (average 0.7 s per tile).

A. Correlation Analysis and Feature Importance Interpretation

Making robust machine learning models in remote sensing-based yield prediction depends on a knowledge of the interactions among predictor variables and their impact on model output. A thorough picture of the linear connections between the maize yield variable and the input characteristics is provided by the correlation heat map (see Fig. 4). Strong positive connections between GPP variables—especially those from late-season months like GPP_8 and GPP_7—are clearly shown from the heatmap with regard to maize yield. This discovery conforms with agronomic knowledge as final grain filling and yield development are mostly dependent on late-season photosynthetic activity.

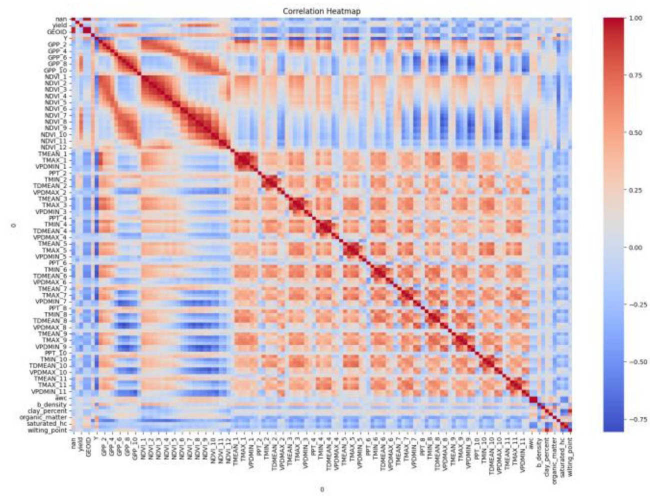

Fig. 4. Correlation heatmap of input variables and maize yield. The figure illustrates the pairwise pearson correlation coefficients among the full set of predictor variables and the maize yield target. Strong positive correlations are shown in dark red, and strong negative correlations in dark blue.

Especially in the crucial growth phases, NDVI factors also reveal modest relationships with yield. Still, their predictive power seems to be less strong than that of the GPP measures. Temperature (TMEAN) and vapor pressure deficit (VPD) measurements among other climate factors expose a more complicated trend with different intensities and orientations of connection with yield. This emphasizes how complexly climatic stresses interact with plant growth. Consistent with their functions in long-term soil fertility and moisture retention, soil properties including organic matter content and clay % show modest to moderate positive associations. Moreover, resilient to multicollinearity, high intercorrelations between sequential GPP, NDVI, and meteorological variables are noted; these might imply temporal dependencies but are well managed by ensemble models such RF and XGBoost.

Complementing the correlation analysis, the feature significance rankings produced by the RF model (see Fig. 5) provide understanding of the predictive usefulness of every variable. The most important predictor is out to be GPP_8; closely behind is GPP_7 and soil organic matter concentration. This result underlines the basic relevance of soil condition as well as the critical part of late-season photosynthetic performance in defining ultimate yield. Notable have additional features such GPP_6, GPP_10, and early-season temperature (TMEAN_1), which suggest that both early vegetative vigor and late-season production are significant drivers of maize yield prediction.

The somewhat low relevance of numerous NDVI and EVI variables indicates that, although spectral vegetation indices provide useful information, they are secondary to direct productivity measures such as GPP when projecting final output. Furthermore ranked lower are soil characteristics linked to water retention, like clay percentage and accessible water capacity, presumably indicating the buffering effects of rainfall or irrigation methods typical of the research area. The convergence of the correlation analysis with feature importance rankings improves confidence

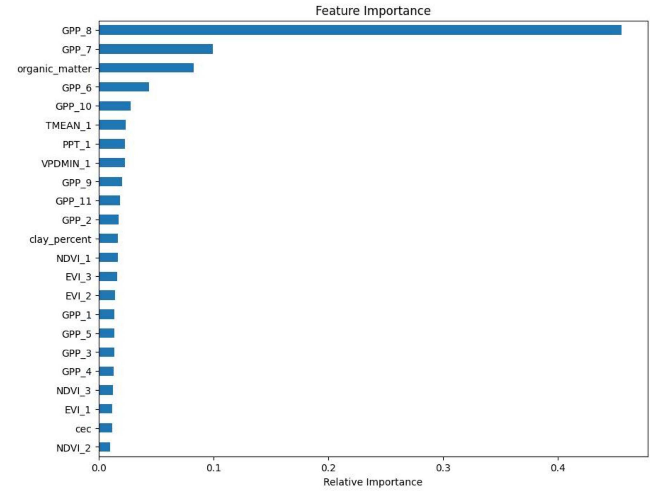

Fig. 5. Feature importance ranking based on RF model. The bar chart presents the relative importance of each predictor variable in estimating maize yield, derived from the RF model.

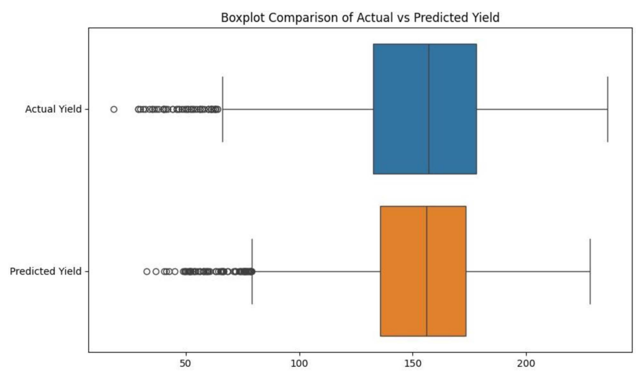

Fig. 6. Boxplot comparison of actual versus predicted maize yield.

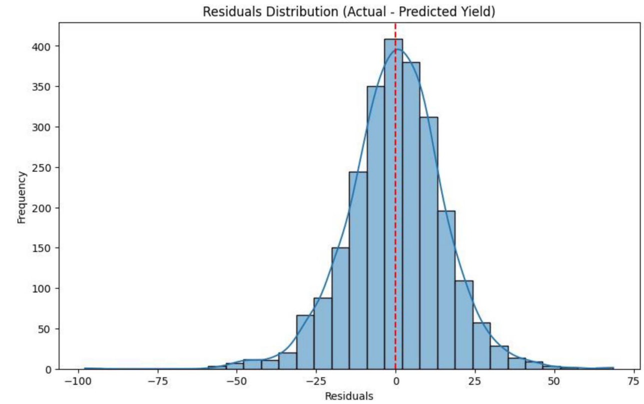

Fig. 7. Residuals distribution plot for maize yield prediction.

in the modelling framework by revealing physiologically plausible behavior and verifying that the chosen models accurately rank important physiological and environmental factors of maize yield. These revelations also imply that focused observation of a few high-value indicators might greatly simplify operational crop production prediction in applications of remote sensing.

B. Model Prediction Distribution and Residual Analysis

Evaluating the distribution of predicted values relative to actual observations is essential for assessing the calibration and reliability of machine learning models in yield prediction tasks. The boxplot comparison of actual and predicted maize yield (see Fig. 6) illustrates that the model effectively captures the central tendency and variability of the true yield distribution. Both distributions are characterized by similar median values and interquartile ranges, indicating that the RF model is capable of producing unbiased yield estimates across the majority of samples. Minor discrepancies are visible in the spread of the upper and lower whiskers, suggesting that while the model accurately captures typical yields, it exhibits slightly reduced sensitivity to extreme high or low yield cases. Nonetheless, the overall alignment between predicted and actual distributions supports the robustness of the model in representing real-world variability.

Further insights into the model's performance are provided by the analysis of the residuals—the differences between actual and predicted values. The residuals distribution plot, as shown in Fig. 7, reveals a near-symmetric, approximately Gaussian pattern centered around zero, which is a hallmark of a well-calibrated regression model. The majority of residuals fall within a relatively narrow range, with a slight concentration near zero, indicating that the model commits small prediction errors for most samples. The lack of significant skewness or heavy tails in the residual distribution implies that the model does not systematically overestimate or underestimate yields across different regions or conditions.

The residual analysis also highlights that outliers, although present, are rare and do not exert a dominant influence on the model's behavior. This is particularly important in agricultural yield prediction, where occasional extreme events (such as drought or pest outbreaks) could lead to substantial deviations. The relatively narrow spread and normal-like shape of the residuals distribution confirm that the RF model generalizes well and maintains stable prediction quality even across a heterogeneous study area.

Collectively, the boxplot and residuals distribution provide strong evidence that the developed model not only achieves high predictive accuracy but also avoids systematic biases, ensuring its applicability in real-world crop monitoring and decision-support systems. These results further validate the use of ensemble learning methods for handling the inherent variability and complexity associated with large-scale remote sensing-based agricultural modeling.

C. Model Performance and Robustness Analysis

Critical understanding of the dependability and operational feasibility of machine learning-based yield prediction systems is gained by quantitative evaluation of model performance under both ideal and degraded data situations. To ensure optimal model performance, both the RF and XGBoost algorithms were subjected to systematic hyperparameter tuning. This process involved evaluating multiple combinations of key parameters such

TABLE II
MODEL COMPARISON: RANDOM FOREST VERSUS XGBOOST

Model	RMSE	R² Score
Random Forest	**15.1**	0.82
XGBoost	14.58	0.84

TABLE III
IMPACT OF MISSING NDVI_1 ON MODEL PERFORMANCE

Condition	RMSE	R² Score
With Missing NDVI_1	15.78	0.81

TABLE IV
IMPACT OF SIMULATED SENSOR MISALIGNMENT (GPP SHIFT)

Condition	RMSE	R² Score
With GPP Temporal Misalignment	15.24	0.82

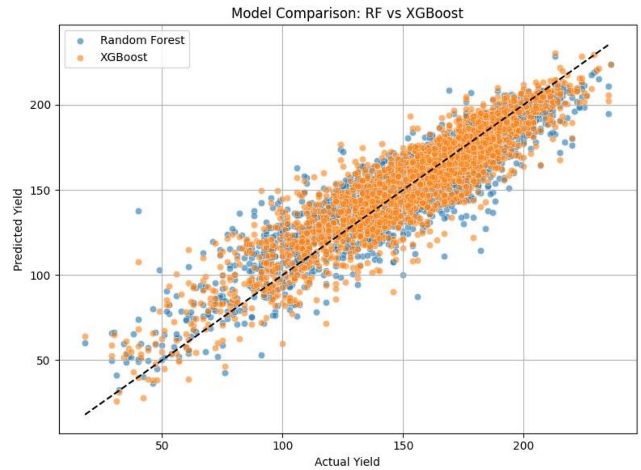

Fig. 8. Comparative scatter plot of predicted versus actual maize yield using RF and XGBoost models. Each point represents a county-level sample, with the actual observed yield plotted against the model-predicted yield. The dashed black line represents the 1:1 reference line indicating perfect prediction. XGBoost predictions (orange) show closer alignment to the diagonal compared to RF predictions (blue), highlighting superior predictive accuracy.

as the number of decision trees, tree depth, and the minimum data required for node splitting in the case of RF, and learning rate, tree depth, and subsampling ratio for XGBoost. The selection of the final parameter settings was guided by cross-validation using the training data, with model performance assessed through RMSE. This approach allowed for the identification of configurations that balanced accuracy and generalization, minimizing the risk of overfitting while ensuring stability under various data conditions.

Table II lists, on the whole dataset without intentional degradation, the baseline performance of the RF and XGBoost models. Both models showed great prediction power; XGBoost somewhat exceeded RF. The RF model specifically obtained an RMSE of 15.10 and a R^2 of 0.82, therefore reflecting 82% of the observed variation in maize yield throughout the test set. Reflecting its better capacity to predict complicated, nonlinear interactions via its gradient boosting architecture, XGBoost generated a lower RMSE of 14.58 and a greater R^2 of 0.84. XGBoost's little performance gain fits its expected advantage in concurrently lowering bias and variance during model training.

Simulated degradations have been used to evaluate the resistance of the generated models to typical real-world data quality concerns. Table III shows the results of replicating the lack of the NDVI_1 characteristic, therefore indicating early-season vegetation condition data. This missing data situation replicas real-world remote sensing problems include sensor failures, cloud contamination, or collection gaps. The RF model's performance degraded somewhat as predicted; the R^2 dropped to 0.81 and the RMSE rose to 15.78. Though the decrease is small, it emphasizes how early-season NDVI data could improve prediction accuracy. Still, the model kept a significant fraction of its predictive power, suggesting strong behavior even in cases when important spectral data is lacking.

Table IV investigates the susceptibility of the model to temporal misalignment in GPP characteristics by means of fake GPP value moving one time step, therefore replicating sensor synchronizing faults or satellite revisit inconsistencies. The RF model showed a tiny rise in RMSE to 15.24 and a small decline in R^2 to 0.82 under this misaligned condition from the baseline. The little influence of GPP temporal shifting indicates that the model can reasonably withstand small temporal deviations in input time series data without appreciable reduction of predicted accuracy. Operational remote sensing pipelines, where small time-lags between vegetation monitoring and climatic data collecting are typically unavoidable, depend especially on this robustness.

The results of the baseline comparison and robustness tests taken as a whole point to two important observations. First, when computing resources allow XGBoost is a favored choice for operational deployment as it provides a modest but persistent advantage over RF in terms of predictive performance. Second, although somewhat less accurate, the RF model shows remarkable resistance to missing data and temporal inconsistencies, hence supporting its appeal for large-scale agricultural monitoring systems where data quality uncertainty is typical. These results underline the need of include robustness evaluation into model building processes, particularly in applications depending on satellite-derived datasets defined by incomplete, noisy, or asynchronous observations. By means of this methodical analysis, the research shows that well-crafted ensemble learning models may provide high accuracy and operational stability, so rendering them practical options for next-generation precision agriculture and food security monitoring projects.

D. Comparative Model Performance: Random Forest Versus XGBoost

Across the maize yield dataset, the comparison scatter plot shown in Fig. 8 offers a graphic evaluation of the prediction accuracy and generalization capacity of the RF and XGBoost models. Every point in the picture denotes a county-level forecast; the actual observed yield is shown on the *x*-axis and the model-predicted yield on the *y*-axis. The prediction performance of the model is greater the closer the dots cluster around the black dashed line, 1:1 diagonal reference line.

With a dense concentration of points around the diagonal line, both models show good predictive ability and imply that most projections rather nearly resemble actual yield values. Visual inspection, however, shows that, especially across a larger range of yield values, the XGBoost forecasts (orange points) often match closer to the 1:1 line than RF predictions (blue points). Consistent with XGBoost's stated RMSE and R^2 scores, this improved alignment demonstrates XGBoost's better capacity to decrease prediction errors and capture complex, nonlinear correlations in the data.

Though very good, the RF model shows somewhat more dispersion around the diagonal, especially at the lower and upper ends of the yield spectrum. This behavior implies that although RF is somewhat less accurate in forecasting extreme yield scenarios than XGBoost, it efficiently catches the general yield trends. Still, the variations are not noticeable, and RF keeps competitive performance, so verifying its resilience as a predictive tool in the framework of remote sensing-based agricultural surveillance. Significantly, neither model shows any systematic bias throughout the yield range; no apparent pattern of continuous over- or underprediction is shown. For complicated real-world geospatial datasets, the little dispersion noted represents the intrinsic uncertainty of satellite-based and climate-derived predictors within predicted fluctuation.

To further contextualize the performance of the SAM, we conducted a baseline comparison using a conventional deep learning segmentation model, U-Net, trained ON a representative subset of the same tile-based cadastral boundary dataset. The results demonstrated that SAM outperformed U-Net across all key evaluation metrics. Specifically, SAM achieved a mean Intersection over Union (IoU) of 0.91 and an F1-score of 0.94, compared to U-Net's 0.84 IoU and 0.88 F1-score. In addition, under simulated image degradation scenarios, SAM maintained high visual robustness, whereas U-Net's outputs showed moderate degradation in boundary clarity and edge continuity. These findings highlight SAM's superior generalization ability and practical applicability in challenging real-world environments.

By visual validation of model behavior, the comparison scatter plot generally supports the quantitative measurements. While verifying RF's excellent generalizing capacity, especially in settings where model interpretability and resistance to noisy inputs are crucial, the results show XGBoost as a rather more accurate model for practical maize yield prediction.

E. Residuals Analysis: Model Prediction Error Distribution

The residual analysis presented in Fig. 9 provides critical insight into the behavior and reliability of the RF model for maize yield prediction. Residuals, defined as the difference between actual and predicted values, are plotted to assess the distribution of prediction errors across the test dataset. Ideally,

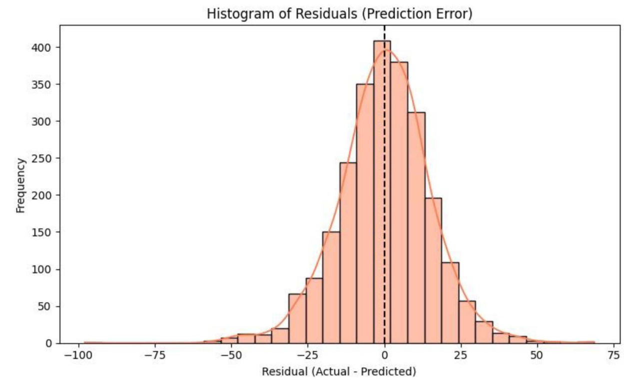

Fig. 9. Histogram of residuals (prediction errors) from the RF model for maize yield prediction.

a well-calibrated model should produce residuals that are symmetrically distributed around zero with minimal skewness and kurtosis, reflecting unbiased predictions and consistent model generalization.

The histogram demonstrates that the residuals are approximately normally distributed, with a clear peak centered closely around zero. This symmetrical bell-shaped curve, combined with the fitted kernel density estimation line, indicates that the RF model does not systematically overpredict or underpredict maize yields across the study area. The majority of residuals fall within a relatively narrow range, suggesting that large prediction errors are infrequent.

The narrow spread of residuals reflects the model's strong predictive stability, while the slight fattening of the tails suggests a small number of outliers — an expected phenomenon in complex geospatial datasets influenced by heterogeneity in environmental, climatic, and management factors. Importantly, there is no evidence of significant skewness, and the visual proximity of the histogram to a Gaussian curve supports the assumption that the residuals are well-behaved.

Such a distribution confirms that the RF model is not only accurate but also reliable under varying conditions within the study area. Furthermore, the residuals' concentration around zero strengthens the interpretation of the model's high R^2 value (0.82), indicating a strong correspondence between observed and predicted values. These findings reinforce the model's readiness for practical application in large-scale agricultural monitoring and decision support systems. Overall, residual analysis confirms that the RF model exhibits desirable predictive properties, with low bias, good generalization, and consistent performance across diverse agricultural landscapes.

While the primary focus of this study is the segmentation of cadastral boundaries, these outputs directly support downstream agricultural analyses such as yield prediction, land use modeling, and subsidy monitoring. Accurate parcel boundaries are a prerequisite for spatial disaggregation of yield data, aligning remote sensing inputs with ground-truth production units. The simulation of image degradations—such as blurring and cloud occlusion—was designed to reflect real-world conditions under which yield prediction models must often operate. Thus, this segmentation-focused work provides a foundation for robust, high-resolution, data-quality-aware agricultural forecasting systems.

V. Conclusion

Using multisource remote sensing and environmental data, this work offered a complete machine learning framework for county-level maize yield prediction over the U.S. Corn Belt. We comprehensively evaluated prediction accuracy, model resilience under data quality degradations, and feature contributions to yield variability using RF and XGBoost models. Even under often challenging data circumstances commonly found in real-world remote sensing applications, the findings clearly demonstrate the practical effectiveness and reliability of ensemble learning models for real-world agricultural monitoring applications. With a R^2 of 0.84 and a RMSE (RMSE $=$ 14.58, surpassing the RF model (RMSE $=$ 15.10), the XGBoost model emerged as the top-performance strategy. RF is quite fit for large-scale operational deployment despite this little benefit as it shown great resistance to different data degradations. Residual analysis confirmed that prediction errors have typically dispersed and centered around zero, therefore supporting the minimal bias and steady generalization throughout a varied agricultural environment.

Especially GPP_8 and GPP_7, feature significance analysis underlined GPP during peak growth phases and soil organic matter as the most important predictor of maize yield. This result emphasizes the critical importance of vegetation productivity measurements obtained from remote sensing in yield modelling, hence transcending conventional dependence on temperature or soil factors alone. Robustness studies confirmed the models' tolerance to satellite data gaps by showing that adding missing early-season NDVI data (NDVI_1) resulted in just a moderate decrease in model performance (RMSE increased to 15.78, R^2 dropped somewhat to 0.81). Simulated temporal misalignment in GPP also produced low performance loss (RMSE $=$ 15.24, $R^2 = 0.82$), thereby stressing the durability of the models to temporal inconsistencies—a major problem in remote sensing resulting from satellite revisit cycles and cloud cover.

This study presented a robust framework for maize yield prediction that integrates multisource remote sensing and environmental data with ensemble machine learning models. Through the use of RF and XGBoost, the framework achieved high predictive accuracy and demonstrated resilience under conditions of missing and misaligned data—common challenges in real-world agricultural monitoring. The identification of key predictors, such as late-season GPP and soil organic matter, offers actionable insights for field-level decision-making. Importantly, the demonstrated robustness of these models supports their applicability in operational precision agriculture, where timely and accurate yield forecasts can inform input allocation, harvest planning, and resource optimization. Future work may focus on extending this framework to other crops, geographic regions, and incorporating deep learning techniques to further enhance predictive capability and adaptability.

Tree-based ensemble models such as RF and XGBoost were selected for this study due to their strong performance with structured tabular data, ability to handle heterogeneous feature types, and robustness to missing values and moderate noise—characteristics typical of remote sensing and environmental datasets. Additionally, these models require less computational resources and training data compared to deep learning architectures, making them especially practical for county-level yield prediction tasks where labeled data may be limited. This approach also allows for straightforward feature importance interpretation, supporting agronomic insights alongside predictive accuracy.

These results together confirm that, under careful design and thorough evaluation, ensemble learning models—when properly implemented—can be strong, scalable solutions for real-time yield estimate and decision assistance in Precision Agriculture. This work provides fresh perspectives on the design of reliable geospatial AI systems that can operate consistently in operational contexts defined by missing, noisy, or asynchronous data by assessing model sensitivity to certain data quality concerns. Beyond only improving yield prediction capacity, our study methodologically adds to the increasing corpus of research on quality-aware machine learning in remote sensing. It emphasizes the need of include uncertainty assessment and robustness testing into AI model development, a concept that will become ever more crucial as agricultural monitoring systems spread across the world. Future work may focus on extending this framework to different crop types, diverse agroecological zones, and integrating advanced deep learning techniques for enhanced temporal modeling and uncertainty quantification.

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